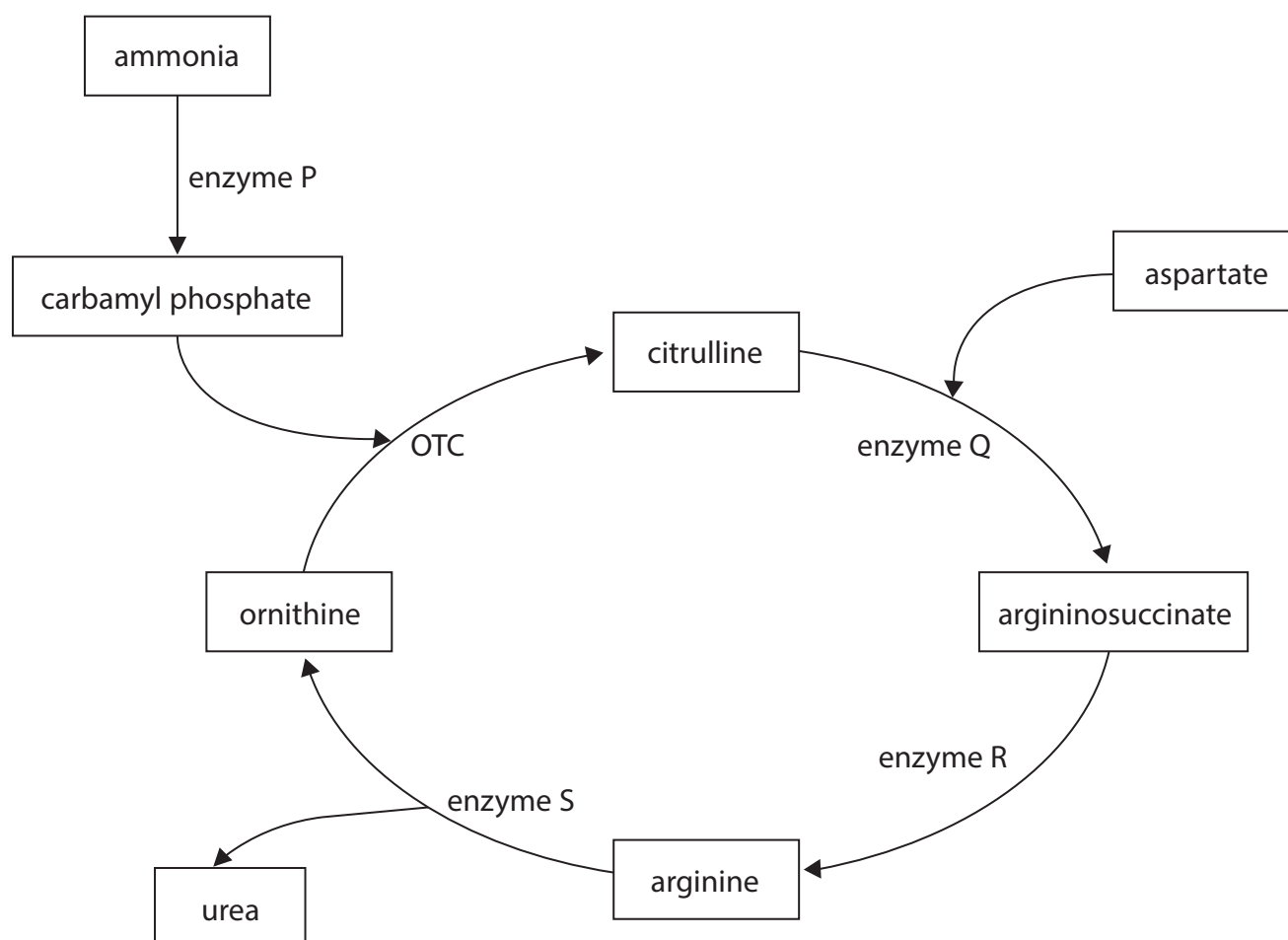


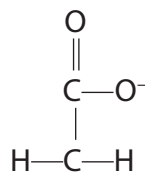
- 8 The amino acid aspartate is converted into the amino acid arginine in the urea cycle.

The urea cycle involves several enzymes and takes place in the liver.

The diagram shows part of the urea cycle.



(a) The R group of aspartate is



Draw the structure of the amino acid aspartate.

(3)

(b) Explain why several enzymes are involved in the urea cycle.

(3)

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(c) Ornithine transcarbamylase (OTC) is one enzyme involved in the urea cycle.

Ornithine transcarbamylase deficiency is an inherited genetic disorder.

(i) Suggest how a person can be shown to have this disorder.

(3)

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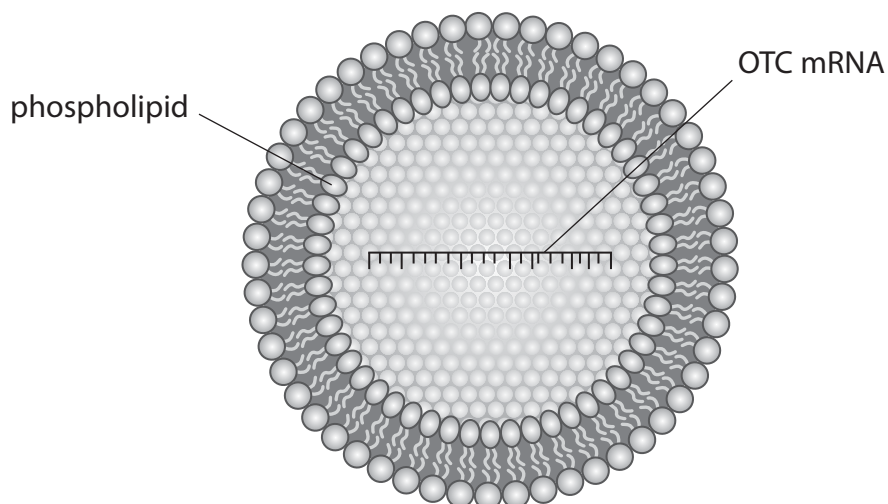
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*(ii) Targeted mRNA therapy is being developed to treat OTC deficiency in mice.

Targeted mRNA therapy involves injecting mice with phospholipid particles containing OTC mRNA.

These particles, shown in the diagram, target liver cells.



The table shows the results of one study using these particles.

Mice	Concentration of ammonia in blood plasma / $\mu\text{mol dm}^{-3}$
normal without treatment	50
with OTC deficiency and without treatment	240
with OTC deficiency and with treatment	40



Explain how this treatment works.

Use the information in the diagram of the urea cycle, the diagram of the particle and the table of results to support your answer.

(6)

DO NOT WRITE IN THIS AREA

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

